

Olist Marketplace EDA

Introduction

E-commerce has transformed the retail industry by enabling businesses to reach customers through online marketplaces while generating valuable data for strategic decision-making. By analyzing transactional, operational, and customer data, organizations can improve sales performance, logistics, and customer satisfaction.

This project explores data from **Olist**, one of Brazil's largest e-commerce marketplaces, which connects thousands of independent sellers with customers across the country. The dataset spans **September 2016 to August 2018** and integrates information on customers, orders, payments, reviews, products, sellers, and geolocation.

During this two-year period, the marketplace recorded:

- **R\$ 15.49 million** in total revenue
- **96,478 customers**
- **96,478 orders**
- **2,970 active sellers**
- **4.08/5 average customer rating**
- Products across multiple categories and Brazilian regions

The objective of this exploratory data analysis (EDA) is to uncover business insights and evaluate marketplace performance using Tableau. The analysis focuses on key areas including **revenue trends, order growth, delivery performance, Pareto analysis of product categories and seller states, seller segmentation**, and the relationship between **delivery experience and customer reviews**. These insights help identify growth opportunities, operational challenges, and the factors that drive customer satisfaction and business performance.

Data Preparation and Methodology

The second phase of this project focused on understanding, preparing, and validating the Olist datasets before conducting the analysis. Each table was carefully reviewed to identify the available information and understand the relationships between customers, orders, order items, payments, reviews, products, sellers, and geolocation data. This process ensured that the dataset contained all the necessary information to answer the project's business questions.

During the data exploration stage, I identified the key fields required for the analysis and assessed the quality and completeness of the data. To prepare the data for visualization, I wrote SQL queries in **Google BigQuery** to extract the required information from multiple related tables. Additional calculated fields were also created to support more detailed analysis and improve the overall reporting capabilities.

SQL Query:

Link: [Capstone_SQL.docx](#)

Following the data extraction process, I performed **seller segmentation** to classify sellers based on their contribution to marketplace performance. This segmentation provides a clearer understanding of seller behavior and enables more meaningful comparisons between different seller groups throughout the analysis.

Seller Segmentation Query:

Link: [Capstone_SQL.docx](#)

Finally, the prepared dataset was imported into **Tableau**, where an interactive dashboard was developed to visualize key performance indicators and answer the project's business questions. The dashboard enables users to explore revenue trends, delivery performance, seller performance, customer behavior, and product category insights through dynamic filters and visualizations.

Tableau Dashboard:

Link: [Dashboard](#)

Analysis & Insights

To support the primary analysis, I developed an interactive **Tableau dashboard** that allows users to explore the data intuitively. Using interactive filters, users can analyze performance across different **time periods, regions, delivery statuses, and seller segments**. This flexibility enables stakeholders to quickly answer key business questions related to sales, revenue, delivery performance, and seller activity without requiring additional data processing.

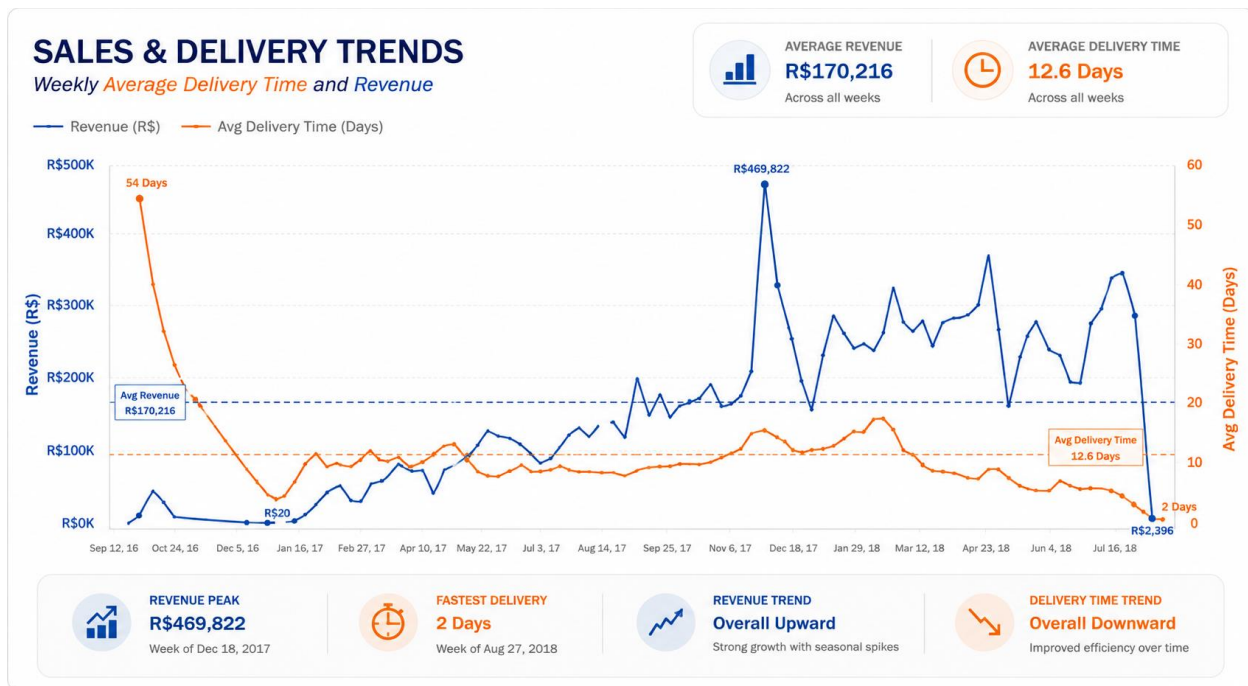
Business Overview

The dashboard begins with a set of key performance indicators (KPIs) that provide a high-level overview of the marketplace's performance during the two-year period. These metrics offer an immediate snapshot of the business, including **total revenue, total orders, active customers, active sellers, and average customer review score**. Presenting these KPIs at the top of the dashboard allows users to quickly understand the overall health of the business before exploring more detailed analyses.



Sales & Delivery Trends

The weekly analysis of sales revenue and average delivery time provides insights into business growth and operational efficiency throughout the analysis period.

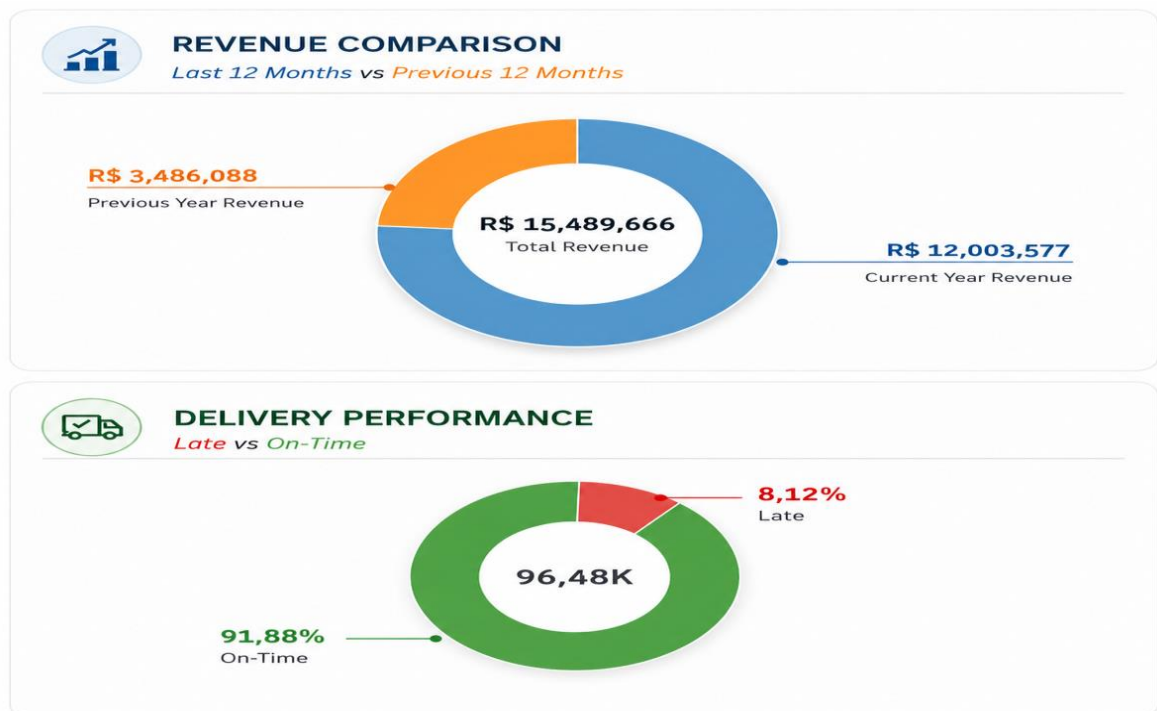


Key Findings

- **Revenue Growth:** Weekly revenue demonstrated a strong upward trend, increasing consistently over time despite periodic fluctuations.
- **Peak Performance:** The highest weekly revenue recorded was **R\$469.8K**, reflecting periods of exceptionally high sales activity.
- **Average Revenue:** The business generated an average weekly revenue of approximately **R\$170.2K**, indicating stable financial performance over the observed period.
- **Delivery Efficiency:** Average delivery time improved significantly, decreasing from **54 days** at the beginning of the period to **2 days** by the end.
- **Average Delivery Time:** The overall average delivery time was **12.6 days**, with a noticeable downward trend throughout the analysis.

Revenue Comparison & Delivery Performance

This analysis compares the company's revenue performance over the most recent 12-month period with the previous 12 months while evaluating delivery performance based on on-time and late deliveries. The results provide insights into business growth and operational efficiency, demonstrating how improvements in logistics supported increased sales and customer satisfaction.



Key Findings

- **Strong Revenue Growth:** Revenue increased from **R\$3.49M** in the previous 12 months to **R\$12.00M** in the last 12 months, representing an increase of approximately **244%**.
- **Total Revenue:** The business generated a cumulative revenue of **R\$15.49M** across the entire analysis period.
- **Current Period Contribution:** The latest 12-month period accounted for approximately **77.5%** of total revenue, indicating significant business expansion.
- **High Delivery Reliability:** **91.88%** of all orders were delivered on time, demonstrating strong logistics performance.
- **Low Late Delivery Rate:** Only **8.12%** of deliveries were completed after the estimated delivery date, reflecting an efficient order fulfillment process.

Revenue by State (Pareto Analysis)

- **São Paulo (SP)** is the largest revenue contributor, generating approximately **one-third of the total revenue**, significantly outperforming all other states.
- Revenue is **highly concentrated** among a small number of states, with **SP, RJ, and MG** contributing the largest share of overall sales.
- Following the **Pareto (80/20) principle**, the **top eight states account for approximately 81% of total revenue**, while the remaining states contribute only around 19%.
- Revenue declines sharply after the leading states, indicating a **long-tail distribution** with many low-performing regions.
- The analysis highlights opportunities to **strengthen sales and marketing efforts in underperforming states** while maintaining strong customer engagement in the highest-revenue markets.



1. REVENUE BY STATE

The revenue by state and revenue distribution in compare with other states.



Top 8 states contribute to ~81% of total revenue, with São Paulo (SP) alone accounting for ~33%.

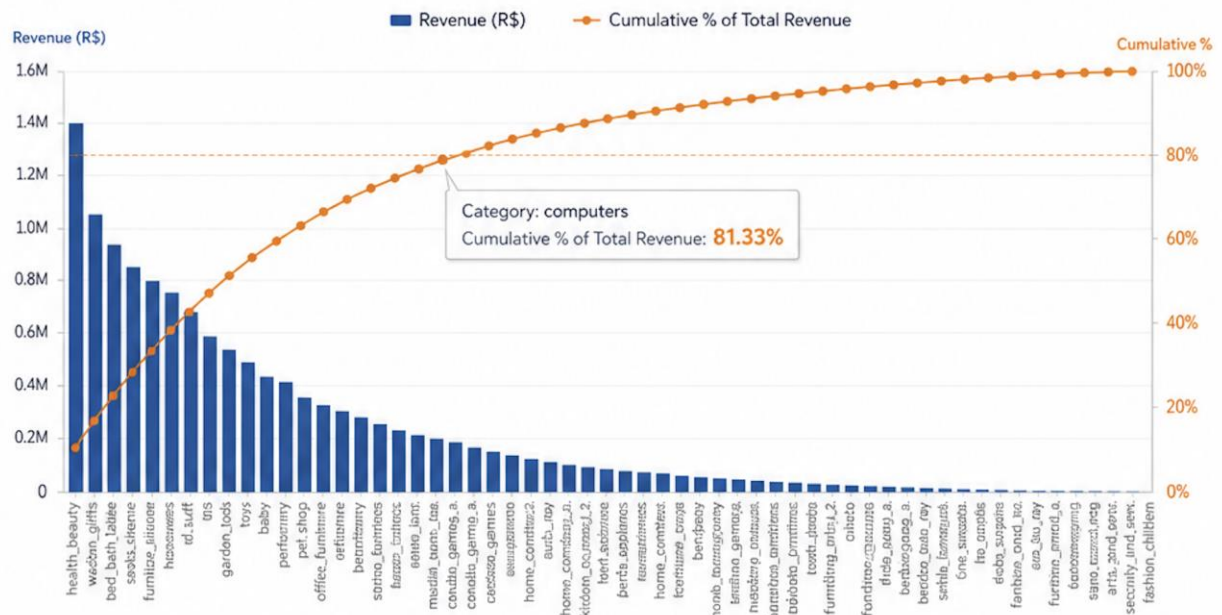
Revenue by Product Category (Pareto Analysis)

- **Health & Beauty** is the highest revenue-generating product category, followed by **Watches & Gifts** and **Bed, Bath & Table**.
- Product revenue follows a **highly skewed distribution**, where a relatively small number of categories generate the majority of sales.
- In line with the **Pareto principle**, the **top 15 product categories contribute approximately 81% of total revenue**, while the remaining categories collectively contribute less than 20%.
- After the leading categories, revenue decreases steadily, indicating a **long-tail portfolio** with many niche product categories generating comparatively lower sales.



2. REVENUE BY PRODUCT CATEGORY

The distribution of product categories with the most revenue and revenue distribution in compare with other product categories.



Top 15 categories contribute to ~81% of total revenue, led by Health & Beauty, Watches & Gifts, and Bed, Bath & Table.

Seller Segmentation

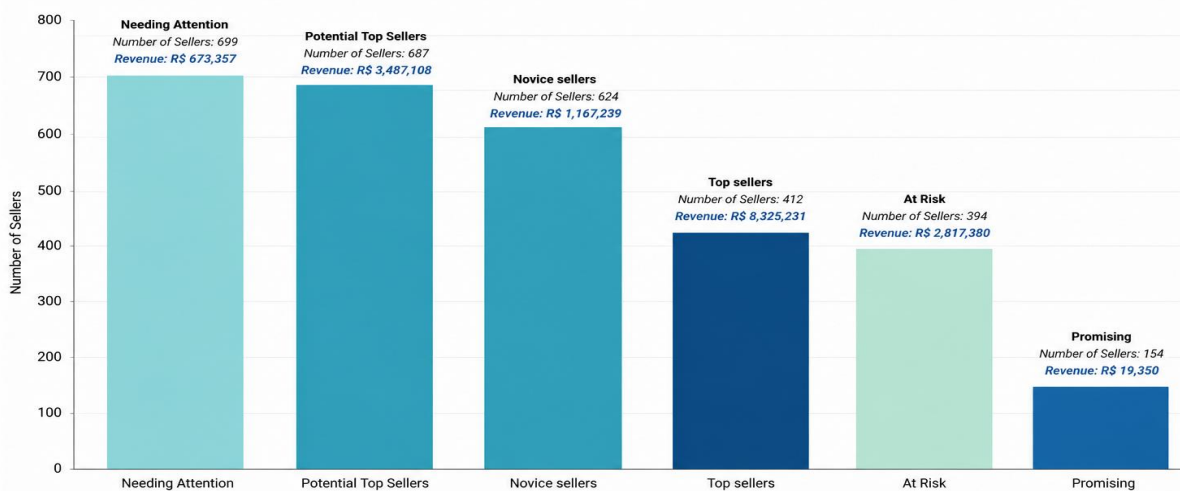
The seller segmentation analysis uses the **RFM (Recency, Frequency, Monetary)** model to classify sellers into six performance groups based on their sales activity, order frequency, and revenue contribution. This helps identify high-performing sellers, sellers with growth potential, and those requiring retention or performance improvement strategies.

Key Insights

- **Top Sellers** generate the highest revenue (**R\$8.33M**) despite representing only **412 sellers**.
- **Potential Top Sellers** (687 sellers) contribute **R\$3.49M**, indicating strong growth potential.
- **Needs Attention** is the largest segment (**699 sellers**) but contributes relatively low revenue (**R\$673K**).
- **At Risk** sellers generate **R\$2.82M**, highlighting the need for retention initiatives.
- **Novice** and **Promising** sellers contribute the least revenue, presenting opportunities for seller development and onboarding.

SELLER SEGMENTATION

Distribution of sellers across segments based on their revenue contribution and number of sellers.



Top Sellers generate the highest revenue (R\$ 8.33M) with 412 sellers, while Needing Attention has the highest number of sellers (699) but contributes the least revenue.

Conclusion & Recommendations

This exploratory data analysis of the Olist Marketplace dataset provided valuable insights into the marketplace's sales performance, operational efficiency, product portfolio, and seller ecosystem between September 2016 and August 2018. By integrating data from multiple business domains and visualizing key metrics in Tableau, the analysis identified the primary drivers of revenue growth as well as opportunities for future business improvement.

Key Conclusions

- The marketplace experienced **strong business growth**, generating **R\$15.49 million** in total revenue, with revenue increasing by approximately **244%** in the most recent 12-month period compared to the previous year.
- Operational performance improved significantly over time, with **average delivery time decreasing from 54 days to only 2 days**, while maintaining an overall **91.9% on-time delivery rate**, demonstrating substantial improvements in logistics.
- Revenue follows the **Pareto Principle**, where a small proportion of states and product categories generate the majority of sales. The **top eight states contribute approximately 81% of total revenue**, while the **top fifteen product categories account for around 81% of overall revenue**.
- The seller segmentation analysis revealed that **Top Sellers generate more than half of the marketplace revenue despite representing a relatively small proportion of sellers**, highlighting the importance of retaining high-performing sellers while developing emerging ones.
- An average customer rating of **4.08 out of 5** indicates generally positive customer satisfaction, suggesting that improvements in delivery performance have contributed to a better overall shopping experience.

Business Recommendations

Based on these findings, several strategic initiatives can further improve marketplace performance:

- **Expand sales in underperforming regions** by investing in localized marketing campaigns, promotional offers, and seller acquisition programs outside the highest-performing states.
- **Strengthen high-performing product categories** while identifying opportunities to grow promising lower-performing categories through targeted promotions and improved product visibility.
- **Retain top-performing sellers** by introducing loyalty programs, performance incentives, and dedicated seller support to protect the marketplace's largest revenue contributors.
- **Develop mid-tier and emerging sellers** through training, marketing support, and operational guidance to increase their long-term contribution and reduce dependence on a small group of sellers.
- **Continue investing in logistics and delivery optimization** to maintain high on-time delivery rates, reduce late deliveries further, and enhance customer satisfaction.
- **Leverage interactive dashboards for ongoing decision-making**, enabling stakeholders to continuously monitor KPIs, identify emerging trends, and respond proactively to changing business conditions.

Overall, this analysis demonstrates how data-driven decision-making can uncover meaningful business insights across sales, logistics, customer experience, and seller performance. The interactive Tableau dashboard provides an effective tool for monitoring marketplace performance and supports strategic decisions aimed at driving sustainable growth, improving operational efficiency, and enhancing customer satisfaction.